

Assignment 2
30 Marks
Deadline: Jul 20, 2026 (Monday)

Question - 1:

Find the RISC-V code of the following machine Codes. **[3.5 + 3.5]**

i. 0x00F29A93

ii. 0x089739B3

Format	Instruction	Opcode	Funct3	Funct7/Funct6
R	ADD	0110011	001	0000000
	SUB	0110011	110	0000001
	AND	0110011	010	0000001
	OR	0110011	011	0000100
	SLL	0110011	101	0000100
I	LD	0000011	000	N/A
	LW	0000011	010	N/A
	LH	0000011	011	N/A
	LB	0000011	001	N/A
	ADDI	0010011	000	N/A
	SLLI	0010011	001	0000000
	SRLI	0010011	010	0000000
	ORI	0010011	011	N/A
S	SD	0100011	000	N/A
	SW	0100011	001	N/A
	SH	0100011	010	N/A
	SB	0100011	011	N/A

Question - 2:

```
while ( (a+13) > (c - 4) ):
    if (a > b and b < c):
        a = b
    elif ( a == 5):
        c = c + 10
    else:
        d = d + 10
```

Construct the equivalent RISC-V code of the above high-level code. Variables a,b,c,d are associated with register X11, X12, X13, X14 respectively **[10]**

Question - 3:

Under which type(s) of instruction do **ORi** and **Store** fall under? **[1+1]**

What is the significance of the funct3 field in case of a Load instruction? **[3]**

Question - 4:

Memory Location	Code	Line Number
	Loop:	
	SLLI X_{10} , X_{22} , 3	
	ADD X_{10} , X_{10} , X_{25}	
	LD X_9 , 0(X_{10})	
	ADDI X_{17} , X_0 , -7	
	BNE X_9 , X_{26} , Exit	
#7048	ADDI X_{22} , X_{22} , 1	
	SD X_{10} , 16(X_{22})	

	BEQ X_0 , X_0 , Loop	
	Exit:	

Find the SB-type instructions from the above code and encode them accordingly. [4 + 4]

Given,

i) opcode = $(103)_{10}$, funct3 = $(011)_2$ opcode for BEQ.

ii) opcode = $(103)_{10}$, funct3 = $(010)_2$ opcode for BNE.

Question - 5: [Ungraded]

```
int hishab(int i, int j) {
    i = i + 8;
    j = j - a[a [0] ];
    while (j < i) {
        j = a [i];
        i = i + 1;
    }
    return j;
}
```

```
k = 2;
l = 3;
a [1] = hishab(k, l);
```

Translate the following codes written in high level language into an instruction sequence written in RISC-V Assembly. Assume that the variables k and l are in registers X5 and X6 respectively. a is a word array, which has its base address stored in X18. You need to use register X5 inside the function in a way so that it does not get overwritten so write your code accordingly.